

Computer Engineering undergraduate with strong foundations in software engineering, data structures & algorithms, artificial intelligence, and natural language processing. Experienced in developing intelligent applications, backend systems, and cloud-based data solutions using Python, SQL, Azure, PyTorch and Hugging Face Transformers. Passionate about building scalable software enhanced by modern AI technologies.

Education

Thapar Institute of Engineering and Technology
B.E. in Computer Engineering (CGPA: 9.50/10)

Patiala, Punjab
August 2023 – May 2027

The Palace School

XII (C.B.S.E) | Science (Percentage: 95%)
X (C.B.S.E) | General (Percentage: 94.83%)

Jaipur, Rajasthan
2022-23
2020-21

Work Experience

Celebal Technologies

June 2026 – August 2026

Data Engineer Intern

Jaipur, Rajasthan

- Processed & analyzed e-commerce datasets using Python, & SQL through data cleaning, feature engineering, & business analytics.
- Conducted end-to-end sales data analysis using SQL, leveraging filtering, aggregations, and business intelligence queries to uncover customer, product, and revenue insights.
- Designed and implemented an automated cloud-based ETL pipeline using Azure Data Factory, Azure Blob Storage, Linked Services, Datasets, and Get Metadata activities for reliable data ingestion and validation.

Projects

Intelligent Posture Detection System | Python, OpenCV, PyTorch, Arduino, FSR Sensors

2026

- Developed real-time posture monitoring system integrating computer vision and pressure sensor data with a dual-mode architecture for reliable posture detection.
- Designed a modular inference pipeline supporting automatic failover between webcam and FSR-based sensing, ensuring uninterrupted monitoring and fault tolerance.
- Implemented an event-driven alert system generating real-time audio notifications and smartphone alerts with corrective posture recommendations for prolonged incorrect sitting.

MedDocX – Clinical Conversation Routing System | Python, PyTorch, Hugging Face Transformers, BART-Large-MNLI, NLP

2026

- Built a transformer-based clinical conversation analysis system using BART-Large-MNLI to classify and route doctor-patient dialogues across multiple medical specialties.
- Developed an NLP pipeline for text preprocessing, semantic classification, and confidence-based decision making using real-world healthcare dialogue datasets with extensibility including speech-to-text integration and voice-assisted healthcare interactions.
- Applied zero-shot learning techniques to conversational AI tasks, enabling scalable workflow automation without requiring specialty-specific model training, and transaction management.

Stock Sentiment Analysis | Python, NLP, Neural Network, PyTorch

2025

- Built a stock market sentiment analysis system to predict market trends using social media posts and financial news headlines.
- Performed Natural Language Processing (NLP) and sentiment classification using neural networks to assess public sentiment and its impact on stock price movement.
- Implemented text preprocessing, feature extraction, and model evaluation to enhance prediction reliability.

Pharmacy Inventory Management System | SQL, Backend Development

2025

- Developed a pharmacy inventory management system to streamline stock control, billing, expiry tracking, and prescription verification.
- Automated GST-compliant billing, real-time stock updates, restock alerts, and expiry notifications, reducing manual errors and improving operational efficiency.
- Enhanced system reliability through data validation, role-based access, and transaction management.

Skills

- Programming Languages:** Python, C, C++, SQL
- Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Scikit-Learn, TensorFlow, PyTorch, Seaborn, Hugging Face Transformers, OpenCV
- Databases:** Oracle SQL, MS SQL Server
- Tools & Platforms:** GitHub, VS Code, MATLAB, AutoCAD, Microsoft Azure (ADF, Blob Storage)
- Strong understanding of arrays, strings, recursion, linked lists, stacks, queues, trees, graphs, and dynamic programming. (200+ LeetCode problems)

Certifications

Accelerating End-to-End Data Science Workflows | NVIDIA

2026

- Built GPU-accelerated data science pipelines using RAPIDS (cuDF, cuML) for data processing, clustering, and machine learning tasks.

Extra Curricular & Achievements

- Gold Medalist, International French Olympiad (Rank 1).
- Captain, Inter-School Cricket Tournament, demonstrating leadership and teamwork.